



FAST National University
of Computer and Emerging Sciences
Compiler Construction (CS-4031)

Quiz-2*: (CLO - 2)(1) Max. Marks: 03

Name: _____

Student ID: _____

INSTRUCTIONS:

- Don't write with Pencil, Even Diagrams should be made with Pen otherwise marks will be deducted.
- Keep answers concise, specific, and directly relevant,

Question 1: You have given following production with non-terminals having attributes, plus a set of semantic rules. You are required to answer the following with reason for each rule. (Marks = 1.5)

- i. Is it consistent with an S-attributed definition?
- ii. Is it consistent with an L-attributed definition?
- iii. Is there any evaluation order at all (i.e., no circular dependency)?

Production: $P \rightarrow QR$

Rules:

- i. $P.s = Q.s + R.s$ (i. **Yes** ii. **Yes** iii. **Yes**)
Reasons: *only synthesized attributes, S-attributed ones also L-Attr. No-circular Dependency*
- ii. $Q.i = P.i$ (i. **NO** ii. **Yes** iii. **Yes**)
Reasons: *contains inherited attributes, so not S-Attr... child Qi is inheriting from parent Q. so valid L-Attr. No-circular Dependency*
- iii. $R.i = Q.s$ (i. **NO** ii. **Yes** iii. **Yes**)
Reasons: *contains inherited attributes so not S-attributed, Ri is inheriting from Left child Q.s so valid L-attributed & no circular Dependency*

Question 2: Given the following SDD that builds an AST and computes a value: (Marks = 1.5)

$E \rightarrow E_1 + T$ { $E.node = \text{mkNode}('+', E_1.node, T.node)$; $E.val = E_1.val + T.val$ }

$E \rightarrow T$ { $E.node = T.node$; $E.val = T.val$ }

$T \rightarrow \text{num}$ { $T.node = \text{mkLeaf}(\text{num.lexval})$; $T.val = \text{num.lexval}$ }

Is this SDD S-attributed? Can it be evaluated in a single bottom-up pass of an LR parser? Explain briefly.

Yes, because every attribute (both node and val) depends only on attributes of child nodes in the parse tree (no inherited attributes).

Yes, because S-attributed SDDs can be evaluated during a bottom-up parse (e.g., LR parsing) by storing attributes on the parser stack and computing them when a production is reduced.



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Production: $M \rightarrow NP$

Rules:

- i. $N.i = M.i$ (i. ~~NO~~ ii. ~~Yes~~ iii. ~~Yes~~)
Reasons: There are inherited attributes so not S, child is inheriting from parent so valid L, and there is no dependency cycle
- ii. $P.i = N.s + P.s$ (i. ~~NO~~ ii. ~~NO~~ iii. ~~NO~~)
Reasons: Depends on its own synth. attribute (P.s), violating L-attr and creating circular dependency
- iii. $M.s = N.s * P.i$ (i. ~~NO~~ ii. ~~Yes~~ iii. ~~Yes~~)
Reasons: ~~only~~ there is inherited attribute so not S-Attributed valid L-Attributed since P.i can inherit from left sibling and NO circular dependency

Question 2: Grammar with embedded actions (simplified SDT):

$E \rightarrow E_1 + T \quad \{ E.val = E_1.val + T.val \}$
 $E \rightarrow T$
 $T \rightarrow num \quad \{ T.val = num.lexval \}$

Input: 3 + 4

Assume an LR parser. Simulate the stack. Show the stack contents (values only) and the reduction step that computes each attribute. (Marks: 1.5)

Input: 3 + 4

Step	Stack (values only)	Action / Reduction	Attribute Computed
1	3	Shift num	—
2	T(3)	Reduce $T \rightarrow \text{num}$	T.val = 3
3	E(3)	Reduce $E \rightarrow T$	E.val = 3
4	E(3) +	Shift '+'	—
5	E(3) + 4	Shift num	—
6	E(3) + T(4)	Reduce $T \rightarrow \text{num}$	T.val = 4
7	E(7)	Reduce $E \rightarrow E_1 + T$	E.val = 3 + 4 = 7

Final Result: E.val = 7



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Question 1: You have given a Grammar with semantic rules

$D \rightarrow T L \quad \{ L.inh = T.type \}$

$T \rightarrow int \quad \{ T.type = integer \}$

$T \rightarrow float \quad \{ T.type = float \}$

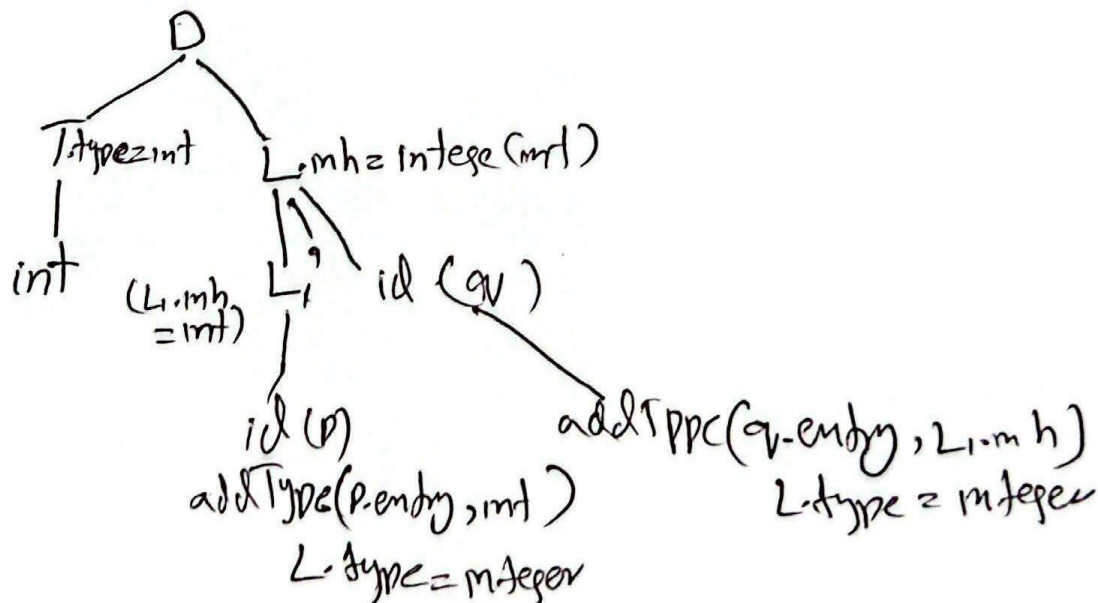
$L \rightarrow L_1, id \quad \{ L_1.inh = L.inh; addType(id.entry, L.inh); L.type = L_1.type \}$

$L \rightarrow id \quad \{ addType(id.entry, L.inh); L.type = L.inh \}$

For the declaration: int p, q

- Draw the parse tree and show both inherited and synthesized attributes.
- Is this SDD L-attributed? Justify your answer with valid reasons
- Can it be evaluated in a single left-to-right traversal? Why or why not?

① Parse tree



② Yes It is L-Inherited Attributed because ^{each} ~~idont~~ ^{idont} ~~per~~ ^{per} inherited from its Left sibling T (type).

③ Yes L-Attributed Definitions can be evaluated during depth first, Left to right traversals (using LL Parser). b/c each inherited attribute depends only on information available from Left siblings.